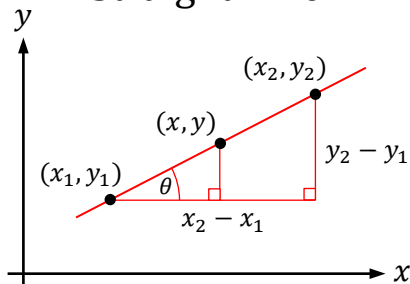


Equation of A Line

Straight Line



$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$



$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} \cdot (x - x_1)$$

$$\frac{y - y_2}{x - x_2} = \frac{y_2 - y_1}{x_2 - x_1}$$



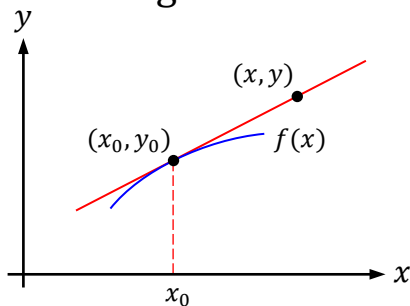
$$y - y_2 = \frac{y_2 - y_1}{x_2 - x_1} \cdot (x - x_2)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{y_1 - y_2}{x_1 - x_2} = \tan \theta$$



$$\begin{aligned} y - y_1 &= m \cdot (x - x_1) \\ y &= mx + (y_1 - mx_1) = mx + c \\ y - y_2 &= m \cdot (x - x_2) \\ y &= mx + (y_2 - mx_2) = mx + c \end{aligned}$$

Tangent Line



$$m = f'(x_0)$$



$$y - y_0 = f'(x_0) \cdot (x - x_0)$$

$$y = 0 : -y_0 = f'(x_0) \cdot (x - x_0)$$

$$x = x_0 - \frac{y_0}{f'(x_0)} = x_0 - \frac{f(x_0)}{f'(x_0)}$$



Newton-Raphson Method

$$\text{Rearranging: } y = y_0 + f'(x_0) \cdot (x - x_0)$$



Explicit Euler Method